CUSTOMER CHURN PREDICTION PROJECT

Zidio Internship Project

Developed by

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Project Duration

One month

(3-11-26 - 3-12-26)

1. Project Introduction

- Project Overview
- Problem Statement
- Project Scope

2. Project Objectives

- Customer Churn Analysis
- Customer Retention Analysis
- Customer Lifetime Value Analysis
- Customer Segmentation
- Churn Risk Identification
- Machine Learning Model Performance Analysis
- Bank Customer Exit Analysis
- Data-Driven Decision Making

3. Architecture Overview

- System Architecture
- Data Flow
- Major Components

4. Project Execution Timeline

- Week 1 – Dataset Collection and Project Planning
- Week 2 – Data Cleaning and Preparation
- Week 3 – DAX Measures and Dashboard Development
- Week 4 – Dashboard Testing, Analysis and Documentation

5. Technical Highlights

- Exploratory Data Analysis
- Power Query
- DAX Measures and Calculated KPIs
- Interactive Power BI Dashboards

- Customer Segmentation
- Churn Risk Analysis
- Machine Learning Model Performance Visualization

6. Dashboard Development

- Customer Churn Analysis Dashboard
- Customer Lifetime Churn Dashboard
- Customer Retention Dashboard
- Customer Segmentation Dashboard
- Customer Churn ML Model Performance Dashboard
- Monthly Charges Churn Dashboard
- Geographic Churn Analysis Dashboard
- Churn Risk Prediction Dashboard
- Bank Exit Analysis Churn Dashboard

7. Results and Analysis

- Key Performance Indicators (KPIs)
- Customer Churn and Retention Insights
- Customer Lifetime Value Insights
- Customer Segmentation Insights
- Geographic Churn Insights
- Churn Risk Insights
- Bank Customer Exit Insights

8. Testing and Validation

- Dataset Validation
- DAX Measure Verification
- Dashboard Visual Validation
- Filter and Slicer Testing
- Results Verification

9. Project Conclusion

- Key Outcomes
- Business Value
- Overall Conclusion

10. Future Enhancements

- Predictive Churn Modeling
- Automated Churn Risk Alerts
- Advanced Customer Segmentation
- Real-Time Customer Churn Monitoring
- Integration of Machine Learning Predictions

11. References

- Datasets Used
- Power BI Documentation
- DAX and Power Query References
- Machine Learning Documentation and Resources

Project Overview:

Customer Churn Analysis is a data analytics and business intelligence project developed to analyze customer behavior, identify churn patterns, understand customer retention, and predict potential customer attrition using data-driven techniques. The project combines Power BI dashboards, Python-based machine learning models, and a Streamlit web application to provide a comprehensive view of customer and banking analytics.

Machine learning models are developed using Python to analyze customer data and evaluate their ability to classify customer churn. Model performance is assessed using evaluation metrics such as accuracy, precision, recall, and F1-score. These metrics help compare model performance and understand how effectively the models identify customers who may leave a service.

Project Objectives:

The primary objective of the Customer Churn Analysis Using Power BI, Machine Learning, and Streamlit project is to analyze customer behavior, identify churn patterns, evaluate machine learning models, and develop interactive dashboards that support data-driven business decisions.

The specific objectives of the project are:

- **Customer Churn Analysis**: Analyze customer data to identify churn rates, churned customers, and factors associated with customer attrition.

- **Customer Retention Analysis**: Examine retained customers and identify patterns that can help businesses improve customer loyalty.

- **Customer Lifetime Value Analysis**: Evaluate customer lifetime value, revenue contributions, and customer-related financial metrics.

- **Customer Segmentation**: Group customers based on relevant characteristics and analyze differences in customer behavior and charges.

- **Churn Risk Identification**: Analyze customer risk indicators to identify customer groups that may require retention efforts.

- **Machine Learning Model Development**: Develop machine learning models using Python to classify customer churn and evaluate their predictive performance.

- **Model Performance Evaluation**: Compare models using evaluation metrics such as accuracy, precision, recall, and F1-score.

Architecture Overview:

The Customer Churn Analysis project integrates data analytics, machine learning, business intelligence, and web-based visualization to analyze customer behavior and identify churn-related patterns. The architecture consists of data collection, data preprocessing, exploratory data analysis, machine learning model development, dashboard creation, and result presentation through Power BI and Streamlit.

The project uses customer and banking datasets to develop analytical dashboards for customer churn, retention, lifetime value, segmentation, churn risk, monthly charges, geographic analysis, and bank customer exit analysis.

System Architecture:

The system architecture describes how raw datasets are processed and transformed into meaningful analytical results through Python, machine learning models, Power BI, and Streamlit.

- **Step 1** — Data Collection: Import customer churn, customer lifetime value, retention, segmentation, monthly charges, geographic, and banking datasets.

- **Step 2** — Data Preparation: Clean and transform the datasets using Python libraries and Power Query.

- **Step 3** — Exploratory Data Analysis: Examine customer distributions, billing patterns, churn behavior, and relevant relationships.

- **Step 4** — Model Development: Prepare model inputs, train machine learning models, and evaluate their performance.

- **Step 5**— Streamlit Integration: Present relevant analytics and machine learning results through the web application.

- **Step 6**— Result Interpretation: Analyze KPIs, dashboard findings, and model evaluation metrics to support business decisions.

Architecture Flow:

Data Sources

↓

Data Cleaning & Preprocessing

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Exploratory Data Analysis & Feature Engineering

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Machine Learning – Customer Churn Prediction

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Machine Learning Model Evaluation

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Power BI Dashboard Development

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Streamlit Web Application Development

↓

Customer Churn Insights & Business Decision Making

Project Execution Timeline:

Project Duration: November 3 – December 3 (One Month)

Week 1 (Nov 3 – Nov 9) – Project Planning and Data Preparation

- Project requirement analysis
- Customer churn dataset collection
- Dataset understanding and column identification
- Data cleaning and preprocessing

Week 2 (Nov 10 – Nov 16) – Exploratory Data Analysis and Feature Engineering

- Exploratory Data Analysis (EDA)
- Customer churn and retention analysis
- Data transformation using Python and Power Query
- Feature engineering and preparation for machine learning

Week 3 (Nov 17 – Nov 23) – Machine Learning Model Development and Dashboard Creation

- Machine learning model training for churn prediction
- Model evaluation using accuracy, precision, recall, and F1-score
- DAX measure development in Power BI
- Development of customer churn, retention, segmentation, and risk dashboards

Week 4 (Nov 24 – Nov 30) – Advanced Analytics and Streamlit Development

- Customer lifetime value and geographic churn analysis
- Monthly charges and bank customer exit analysis
- Machine learning model performance dashboard
- Streamlit web application development and integration of analytical results

Final Phase (Dec 1 – Dec 3) – Testing and Documentation

- Dashboard testing and validation
- Verification of DAX measures and model evaluation results
- Streamlit application testing
- Final project documentation and conclusion

Machine Learning Models Used:

The following models used for customer churn classification.

- **Logistic Regression**: A classification algorithm used to estimate the probability that a customer will churn.
- **Decision Tree Classifier**: Identified patterns in customer attributes and generates rule-based churn predictions.
- **Random Forest Classifier**: Combined multiple decision trees to classify customers and capture complex relationships.
- **Support Vector Machine:** Separated customer records into churn and non-churn classes using a decision boundary.
- **Gradient Boosting Classifier**: Built sequential models to improve classification performance by learning from previous errors.
- **Model performance**: Evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The models' actual scores should be reported from your evaluation results.

Streamlit Dashboard and Its Features:

- **Customer Overview** – Presents key customer statistics and business performance indicators.
- **Churn Analysis** – Analyzes churned and retained customers and identifies churn patterns.
- **Customer Segmentation** – Examines customer groups based on relevant customer attributes.
- **Monthly Charges Analysis** – Visualizes monthly charges and their relationship with customer churn.

- **Geographic Churn Analysis** – Examines churn patterns across geographic locations.
- **Churn Risk Analysis** – Displays customer churn risk indicators.
- **Machine Learning Model Performance** – Presents model evaluation metrics, such as accuracy, precision, recall, and F1-score, when available.
- **Interactive Data Visualizations** – Uses charts, graphs, and KPI cards to communicate analytical results.
- **Data Filtering and Exploration** – Enables users to explore relevant customer records and categories, depending on the implemented filters.
- **Business Insights** – Helps identify potential customer retention opportunities through analytical results.

Power BI Dashboard and Its Features:

- **Customer Churn Analysis Dashboard** – Analyzes customer churn patterns and churn-related KPIs.
- **Customer Lifetime Churn Dashboard** – Examines customer lifetime value and customer churn behavior.
- **Customer Retention Dashboard** – Tracks retained customers and customer retention trends.
- **Customer Segmentation Dashboard** – Analyzes customer groups based on relevant attributes.
- **Customer Churn ML Model Performance Dashboard** – Displays machine learning model evaluation metrics.
- **Monthly Charges Churn Dashboard** – Analyzes monthly charges and their relationship with churn.
- **Geographic Churn Analysis Dashboard** – Visualizes customer churn patterns across geographic locations.
- **Churn Risk Prediction Dashboard** – Presents churn risk indicators and prediction-related insights.
- **Bank Exit Analysis Churn Dashboard** – Analyzes bank customer exits and customer characteristics.
- **Key Performance Indicators (KPIs**) – Displays important metrics through cards and visual summaries.

- **DAX Measures** – Calculates business metrics such as churn rate, customer counts, and average charges, depending on the dataset.
- **Power Query Data Transformation** – Supports data cleaning and preparation before visualization.
- **Interactive Filters and Slicers** – Enables users to explore data by available categories and customer attributes.
- **Business Insights and Reporting** – Helps identify churn patterns and potential customer retention opportunities.

DAX Measures and Formulas Used in Customer Churn Analysis:

❖ Total Customers = DISTINCTCOUNT('Churn'[CustomerID])

❖ Churned Customers = CALCULATE(DISTINCTCOUNT('Churn'[CustomerID]), 'Churn'[Churn] = "Yes")

❖ Retained Customers = CALCULATE(DISTINCTCOUNT('Churn'[CustomerID]), 'Churn'[Churn] = "No")

❖ Churn Rate (%) = DIVIDE([Churned Customers], [Total Customers], 0) * 100

❖ Retention Rate (%) = DIVIDE([Retained Customers], [Total Customers], 0) * 100

❖ Total Monthly Charges = SUM('Churn'[MonthlyCharges])

❖ Average Monthly Charges = AVERAGE('Churn'[MonthlyCharges])

❖ Total Charges = SUM('Churn'[TotalCharges])

❖ Average Tenure = AVERAGE('Churn'[Tenure])

❖ Maximum Monthly Charges = MAX('Churn'[MonthlyCharges])

❖ Minimum Monthly Charges = MIN('Churn'[MonthlyCharges])

❖ Total CLV = SUM('CLV'[CLV])

❖ Average CLV = AVERAGE('CLV'[CLV])

❖ Maximum CLV = MAX('CLV'[CLV])

❖ Minimum CLV = MIN('CLV'[CLV])

❖ Model Recall = AVERAGE('ModelResults'[Recall])

❖ Exited Customers = CALCULATE(DISTINCTCOUNT('Bank'[CustomerId]), 'Bank'[Exited] = "Yes")

❖ Active Customers = CALCULATE(DISTINCTCOUNT('Bank'[CustomerId]), 'Bank'[Exited] = "No")

❖ Customer Exit Rate (%) = DIVIDE([Exited Customers], [Total Bank Customers], 0) * 100

Tools and Technologies Used in Customer Churn Analysis Project:

1)Python - Used as the main programming language for data analysis, preprocessing, exploratory data analysis, and machine learning model development.

2. Pandas and NumPy - Used for loading, manipulating, cleaning, and processing customer churn, customer lifetime value, and banking datasets.

3. Matplotlib and Seaborn - Used to create charts, distribution plots, and correlation heatmaps for exploratory data analysis and customer behavior visualization.

4. Scikit-learn - Used for machine learning model development, data preprocessing, classification, model training, and performance evaluation for customer churn prediction.

5. Google Colab - Used as development environments for executing Python code, analyzing datasets, training machine learning models, and documenting the analytical workflow.

6. Streamlit - Used to develop an interactive web application for presenting customer churn analytics, visualizations, and relevant machine learning model results.

7. Power BI - Used for business intelligence reporting, interactive dashboard development, KPI visualization, and customer churn analysis using DAX measures.

8. Power Query - Used for data cleaning, transformation, column formatting.

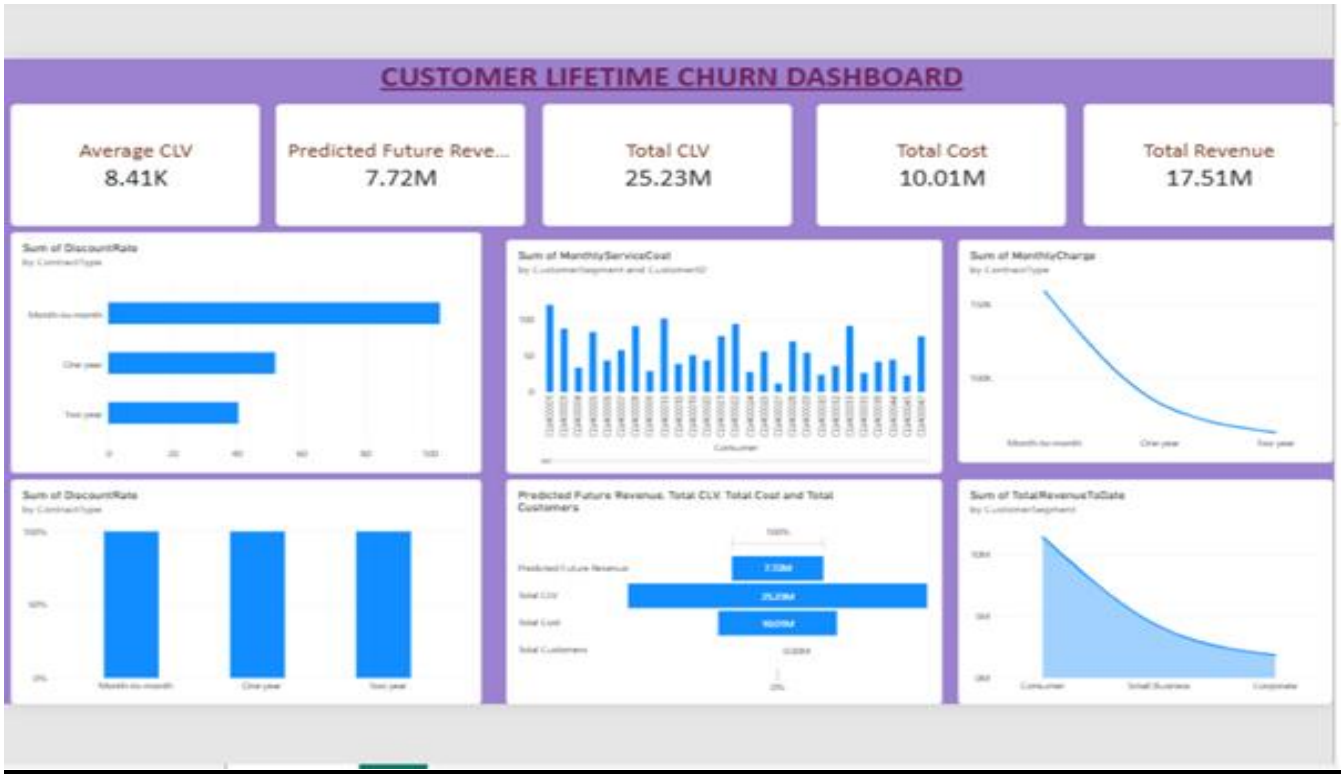
9. DAX (Data Analysis Expressions) - Used to calculate churn rate, retained customers, customer counts, average monthly charges, customer lifetime value, and other business performance indicators.

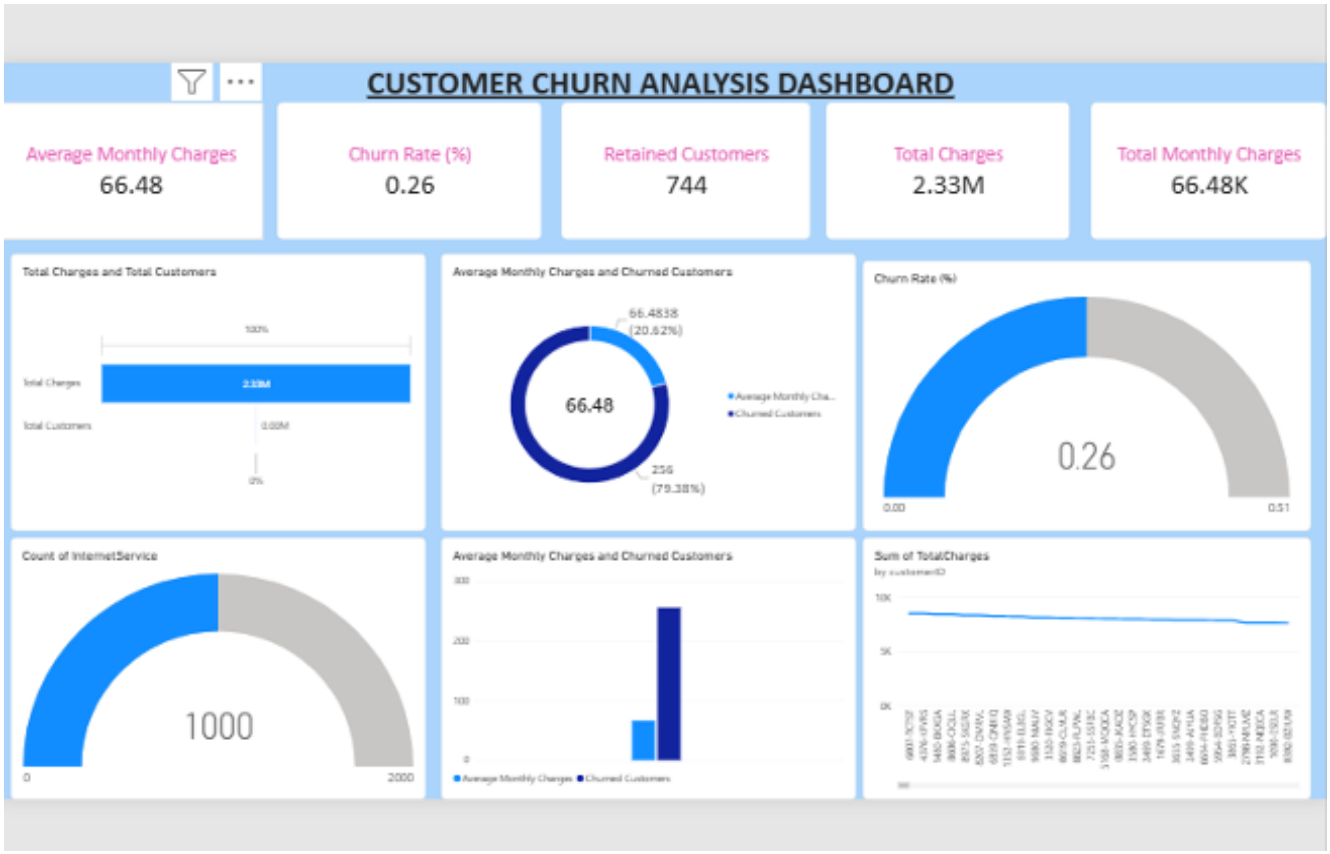
10. Microsoft Excel - Used for storing, inspecting, organizing, and preparing datasets before data analysis and dashboard development.

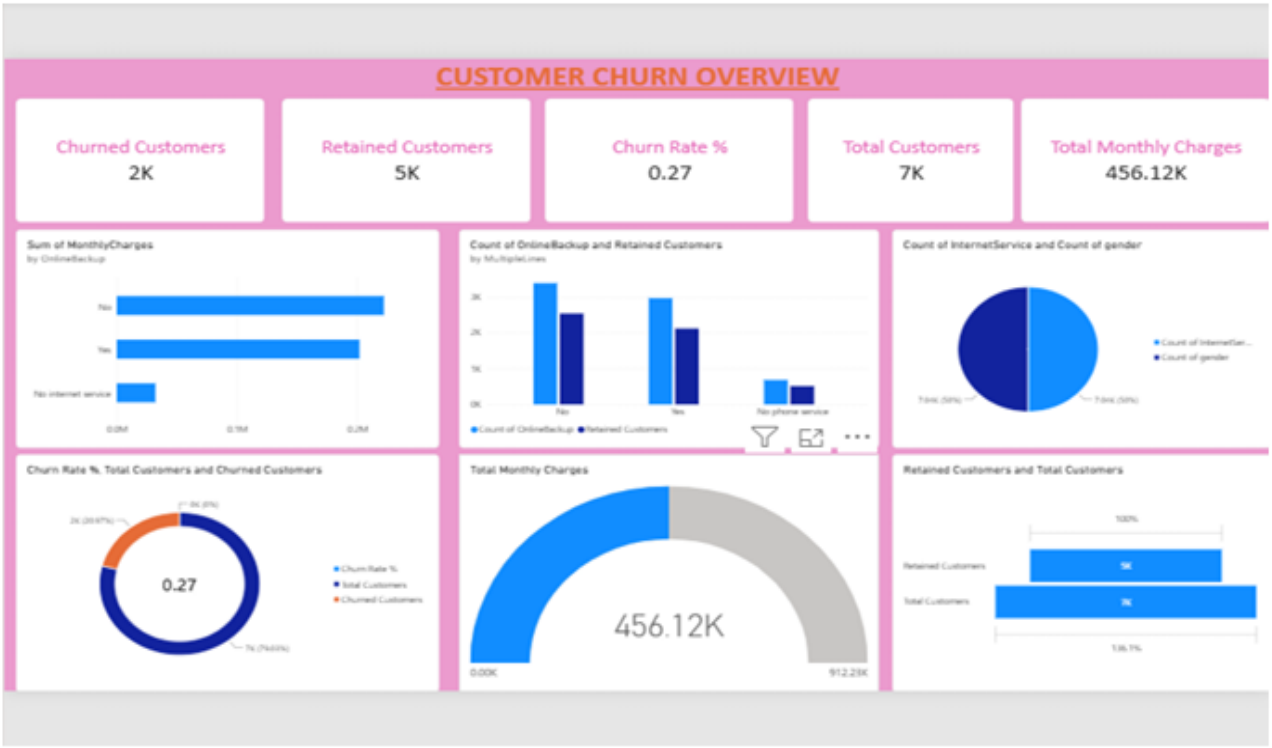
11. GitHub - Used for version control and storing project source code, datasets, and related project resources, where applicable.

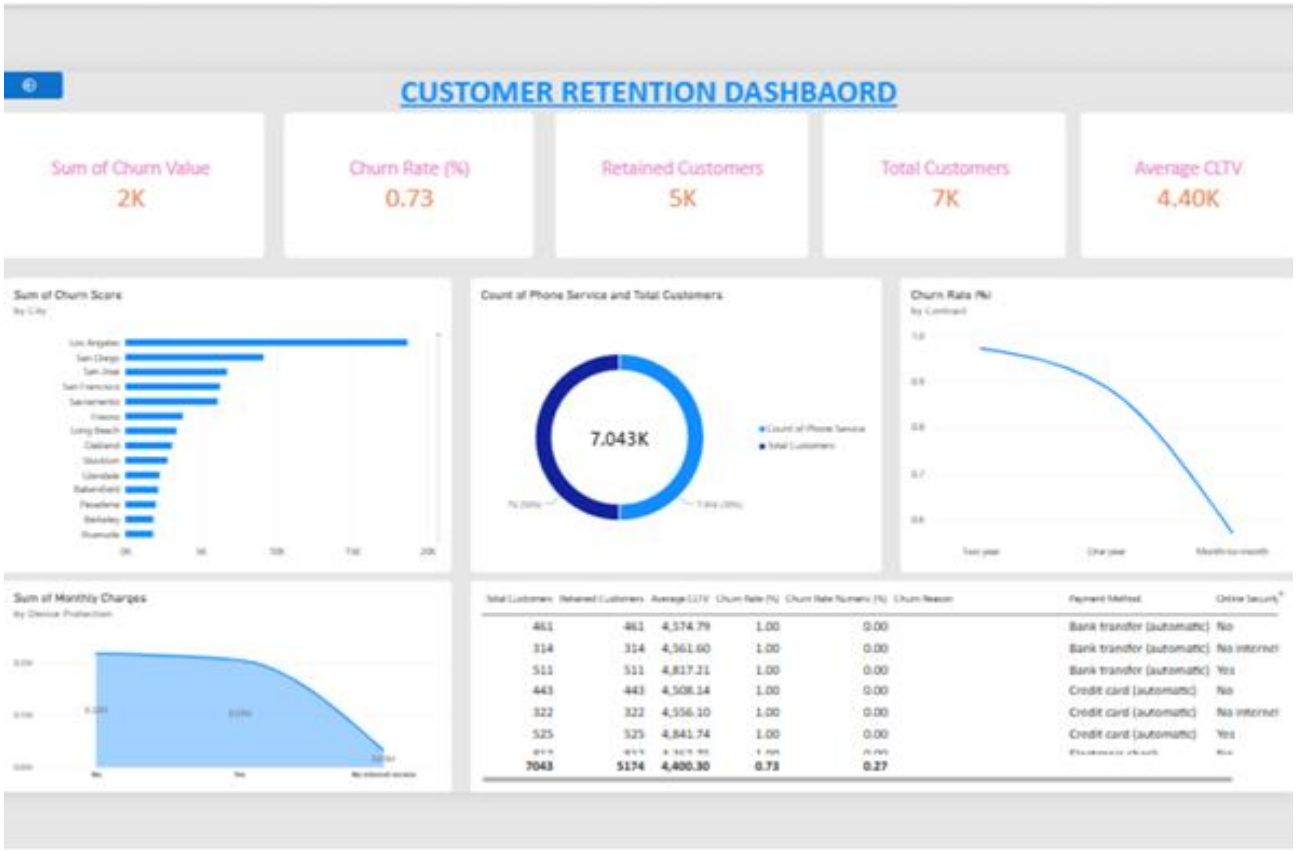
12. Machine Learning Algorithms - Used to develop and evaluate customer churn classification models and analyze metrics such as accuracy, precision, recall, and F1-score.

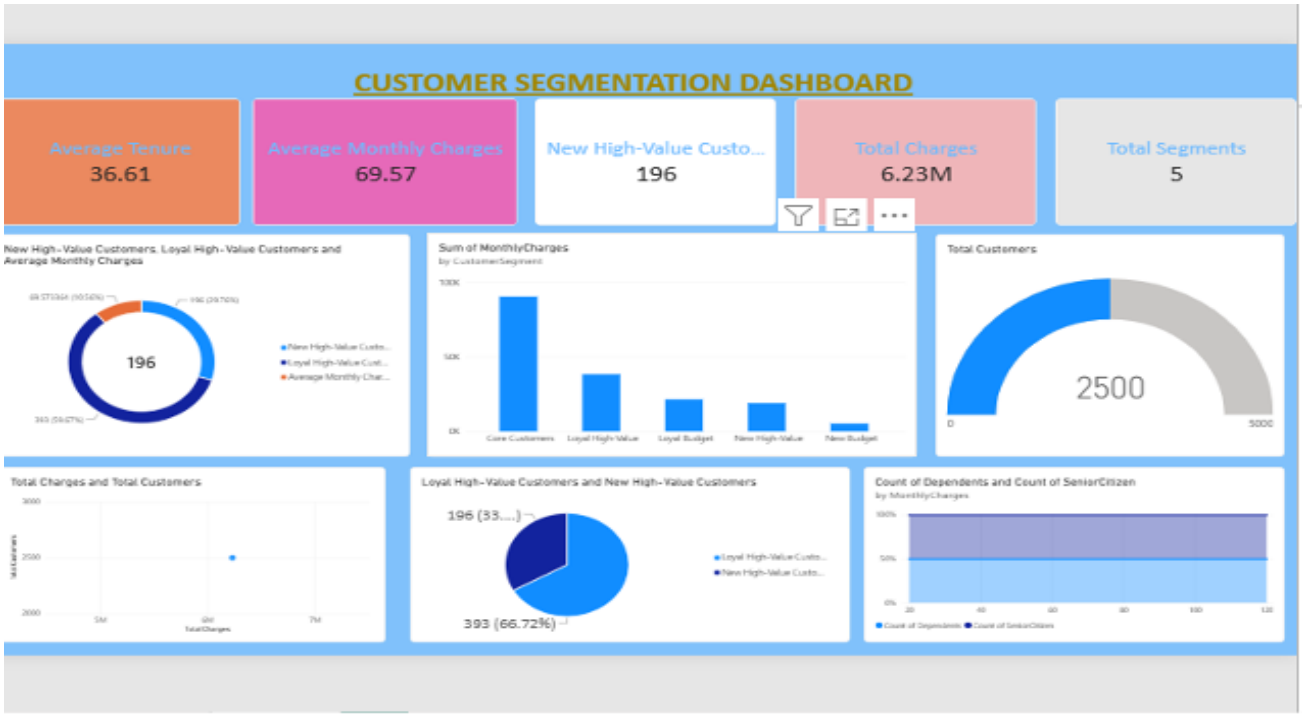
POWERBI DASHBOARD VISUALS:

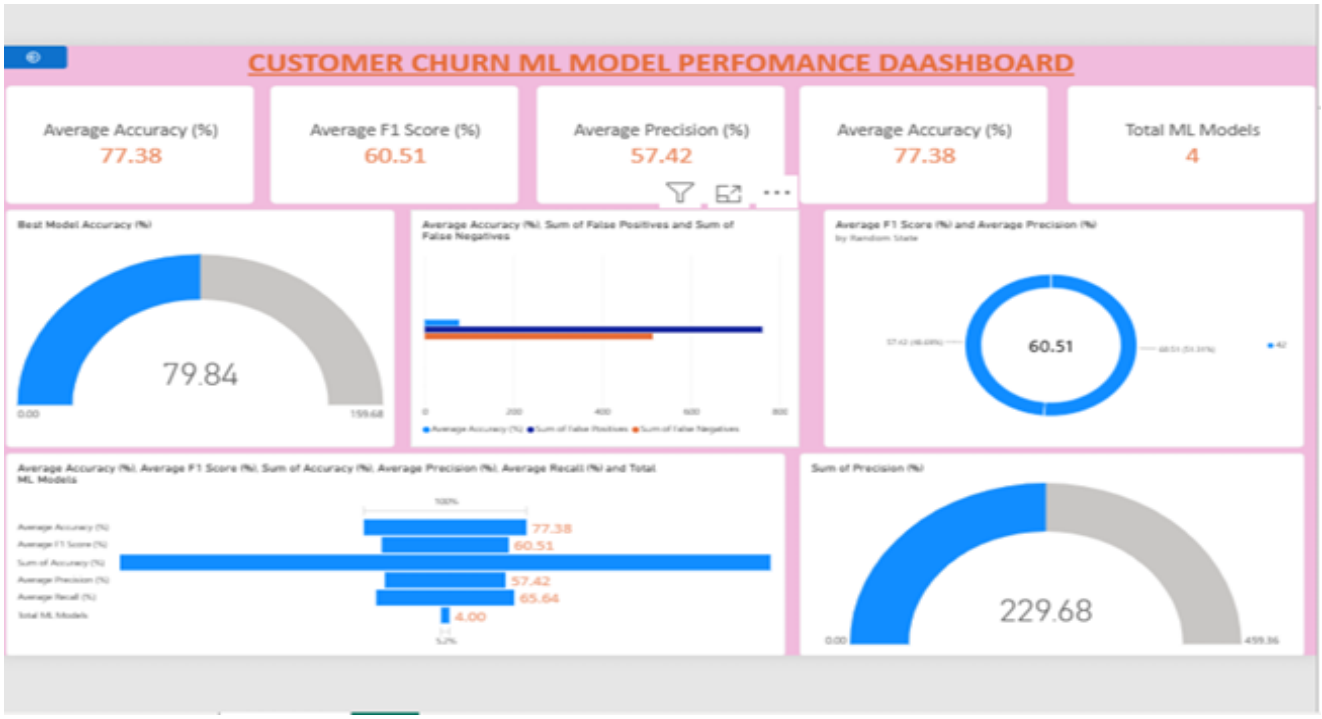


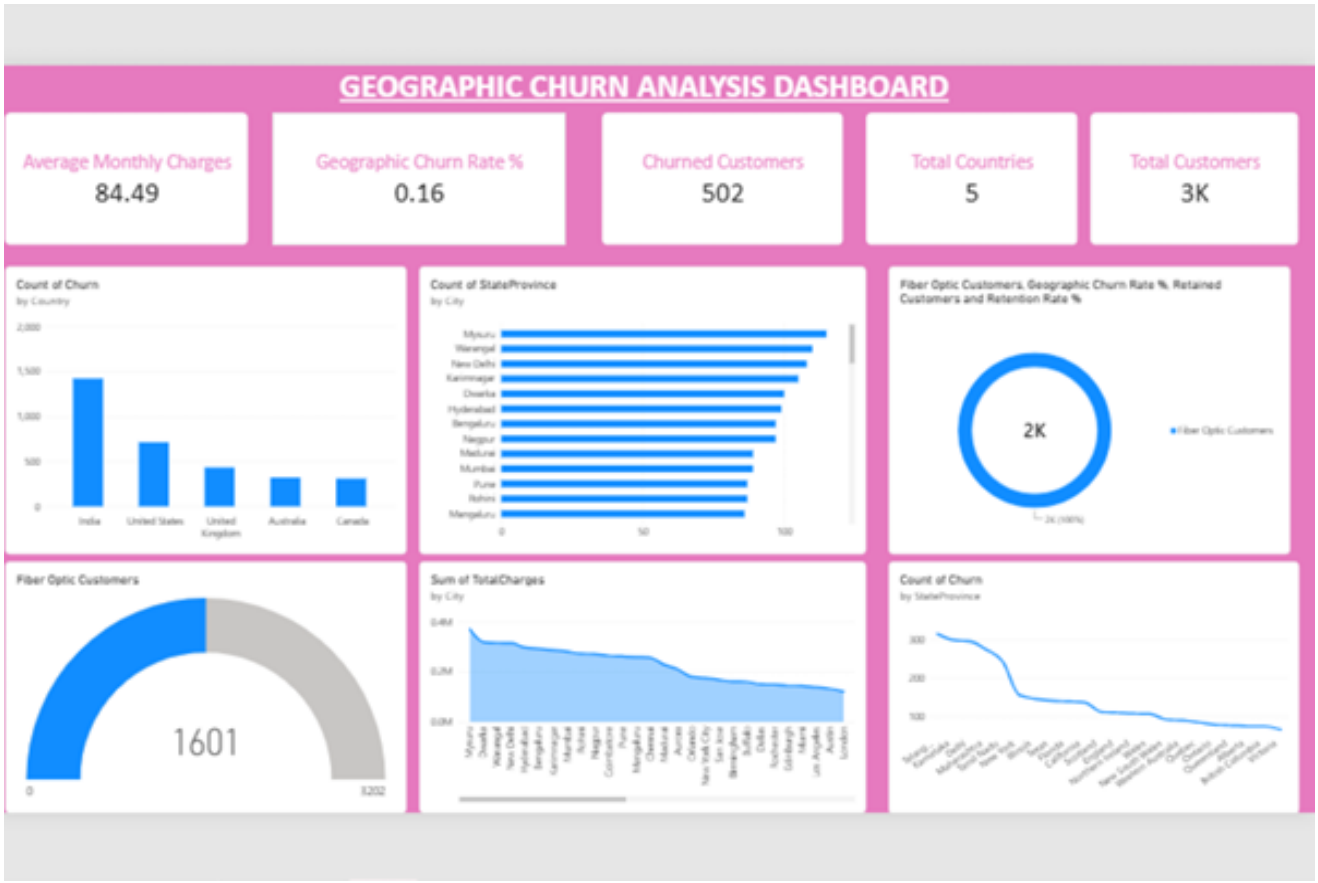

CUSTOMER CHURN ANALYSIS DASHBOARD
Average Monthly Charges
66.48
Churn Rate (%)
0.26
Retained Customers
744
Total Charges
2.33M
Total Monthly Charges
66.48K
Total Charges and Total Customers
Average Monthly Charges and Churned Customers
66.48
Churn Rate (%)
0.26
Count of InternetService
1000
Average Monthly Charges and Churned Customers
Sum of TotalCharges

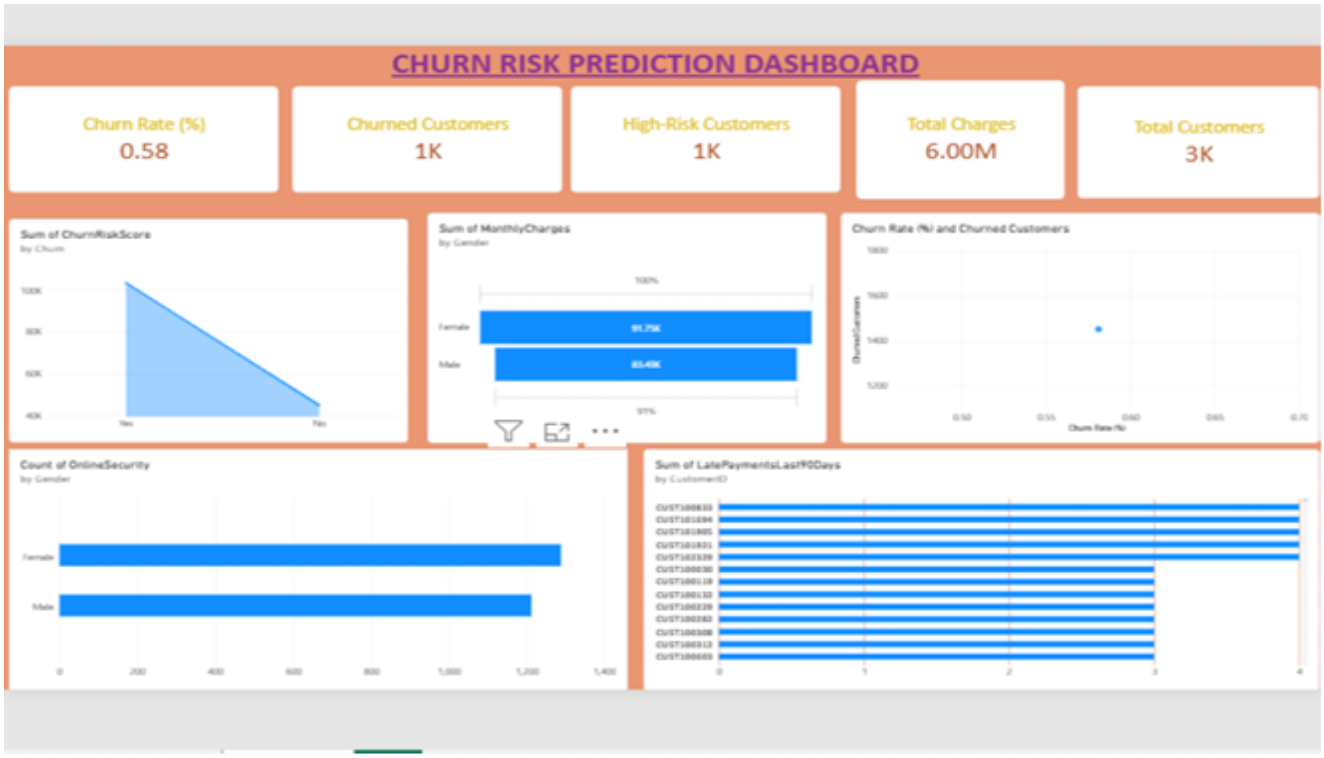

CUSTOMER CHURN OVERVIEW
Churned Customers
2K
Retained Customers
5K
Churn Rate %
0.27
Total Customers
7K
Total Monthly Charges
456.12K
Sum of MonthlyCharges
Count of OnlineBackup and Retained Customers
Count of InternetService and Count of gender
Churn Rate %, Total Customers and Churned Customers
0.27
Total Monthly Charges
456.12K
Retained Customers and Total Customers

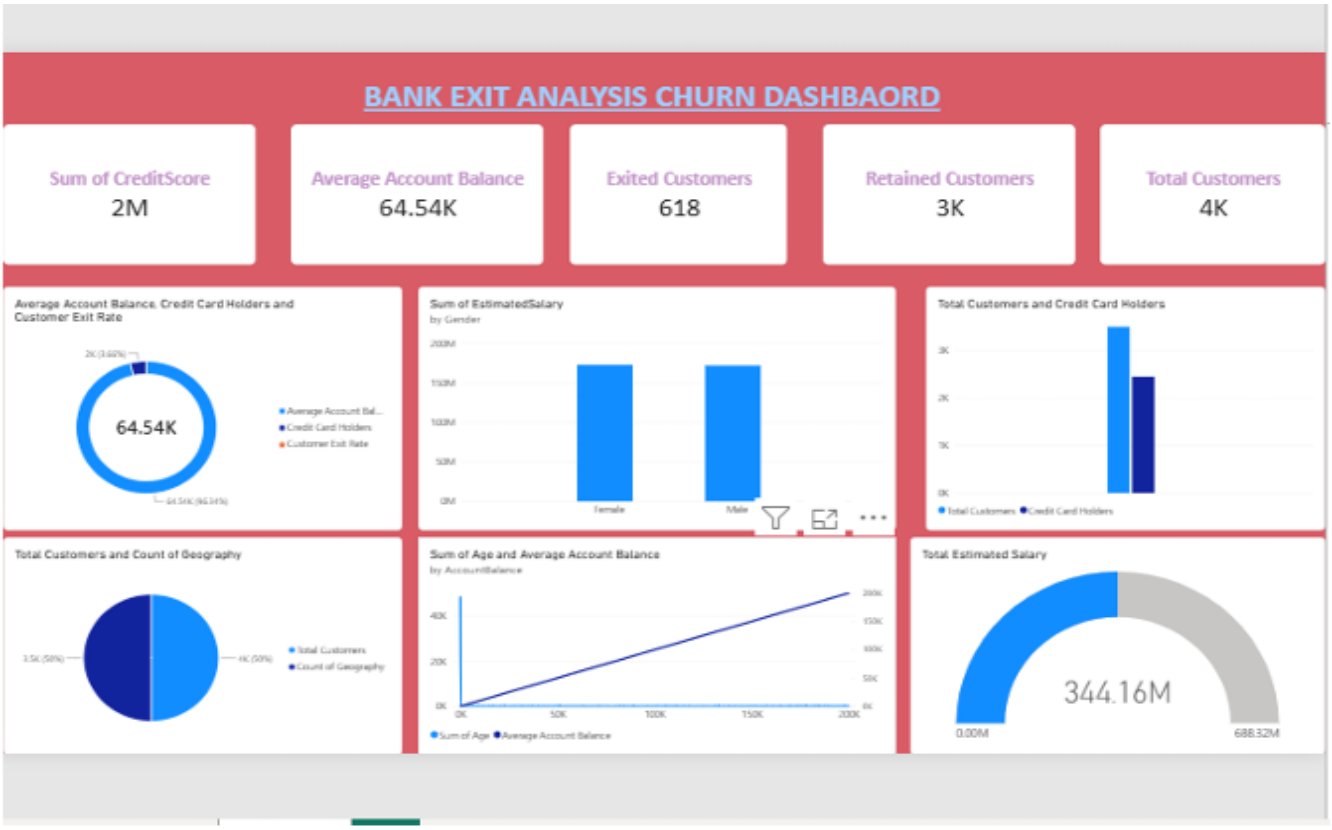

CUSTOMER RETENTION DASHBAORD
Sum of Churn Value
2K
Churn Rate (%)
0.73
Retained Customers
5K
Total Customers
7K
Average CLTV
4.40K
Sum of Churn Score
Count of Phone Service and Total Customers
7.043K
Churn Rate (%)
Sum of Monthly Charges
461 461 4,574.79 1.00 0.00 Bank transfer (automatic) No
314 314 4,561.60 1.00 0.00 Bank transfer (automatic) No internet
511 511 4,817.21 1.00 0.00 Bank transfer (automatic) Yes
443 443 4,508.14 1.00 0.00 Credit card (automatic) No
322 322 4,556.10 1.00 0.00 Credit card (automatic) No internet
525 525 4,841.74 1.00 0.00 Credit card (automatic) Yes
7043 5174 4,400.30 0.73 0.27


CUSTOMER SEGMENTATION DASHBOARD
Average Tenure
36.61
Average Monthly Charges
69.57
New High-Value Custo...
196
Total Charges
6.23M
Total Segments
5
New High-Value Customers, Loyal High-Value Customers and Average Monthly Charges
196
Sum of MonthlyCharges
Total Customers
2500
Total Charges and Total Customers
Loyal High-Value Customers and New High-Value Customers
196 (33....)
393 (66.72%)
Count of Dependents and Count of SeniorCitizen

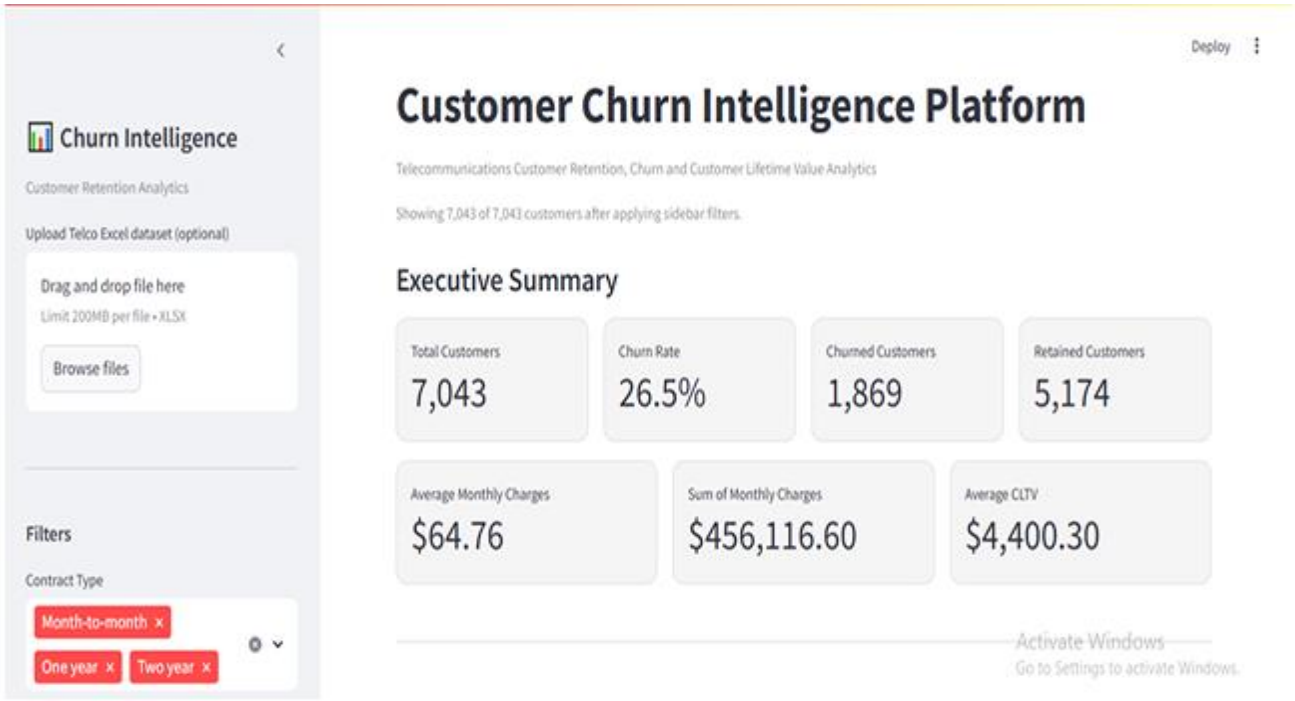

CUSTOMER CHURN ML MODEL PERFOMANCE DAASHBOARD
Average Accuracy (%)
77.38
Average F1 Score (%)
60.51
Average Precision (%)
57.42
Average Accuracy (%)
77.38
Total ML Models
4
79.84
60.51
77.38
60.51
57.42
65.64
4.00
229.68


GEOGRAPHIC CHURN ANALYSIS DASHBOARD
Average Monthly Charges
84.49
Geographic Churn Rate %
0.16
Churned Customers
502
Total Countries
5
Total Customers
3K
Count of Churn
by Country
India
United States
United Kingdom
Australia
Canada
Count of StateProvince
by City
Fiber Optic Customers, Geographic Churn Rate %, Retained Customers and Retention Rate %
2K
Fiber Optic Customers
1601
Sum of TotalCharges
by City
Count of Churn
by StateProvince

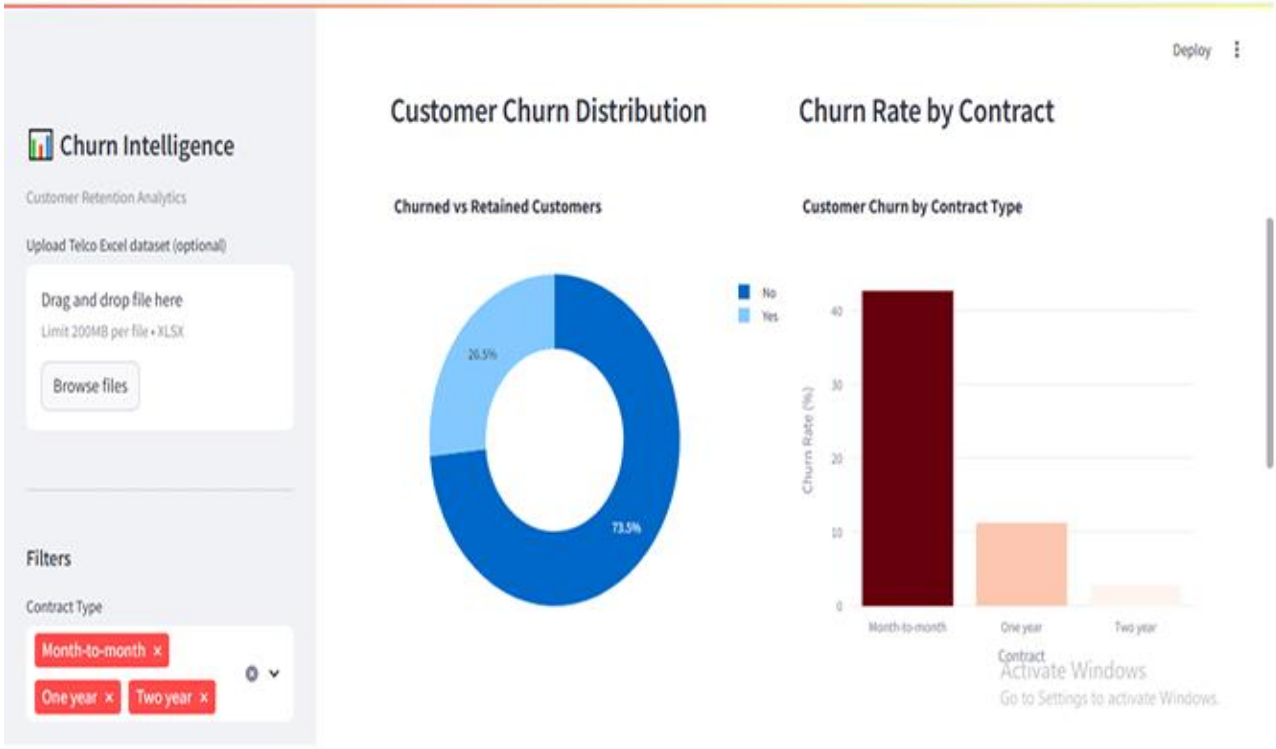

CHURN RISK PREDICTION DASHBOARD
Churn Rate (%)
0.58
Churned Customers
1K
High-Risk Customers
1K
Total Charges
6.00M
Total Customers
3K
Sum of ChurnRiskScore
Sum of MonthlyCharges
Churn Rate (%) and Churned Customers
Count of OnlineSecurity
Sum of LatePaymentsLast90Days

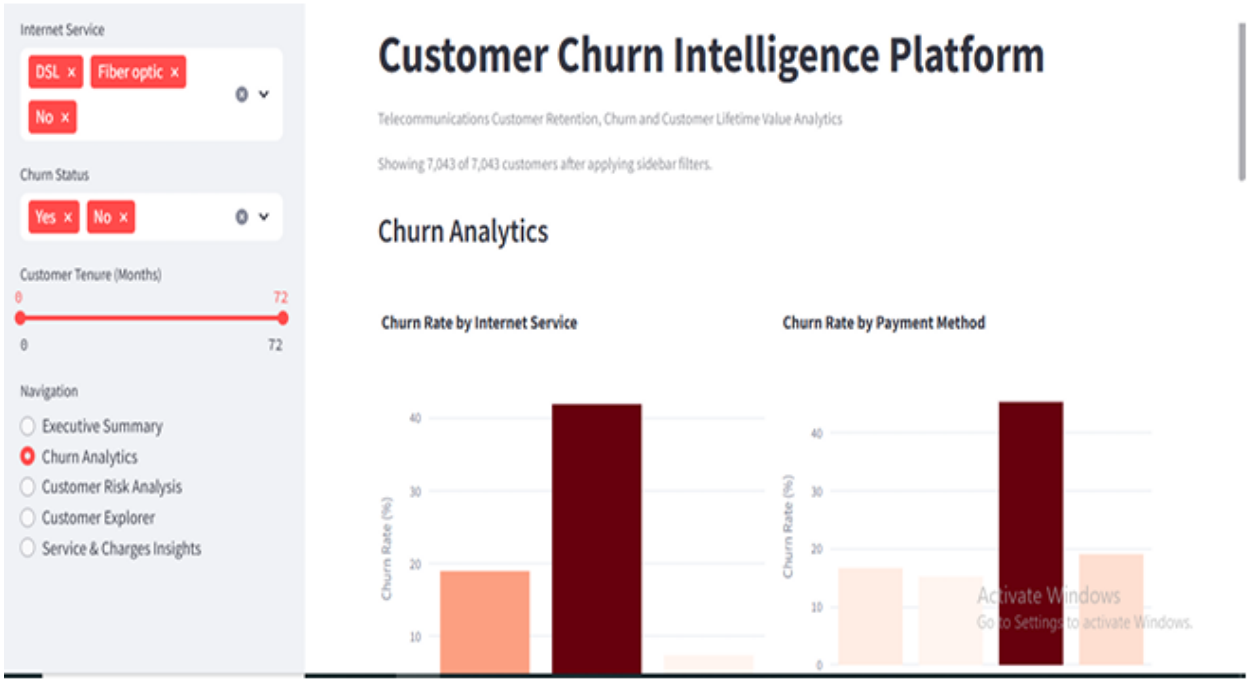

BANK EXIT ANALYSIS CHURN DASHBAORD
Sum of CreditScore
2M
Average Account Balance
64.54K
Exited Customers
618
Retained Customers
3K
Total Customers
4K
Average Account Balance, Credit Card Holders and Customer Exit Rate
64.54K
Sum of EstimatedSalary
Total Customers and Credit Card Holders
Total Customers and Count of Geography
Sum of Age and Average Account Balance
Total Estimated Salary
344.16M

STREMALIT DASHBOARD VISUALS:

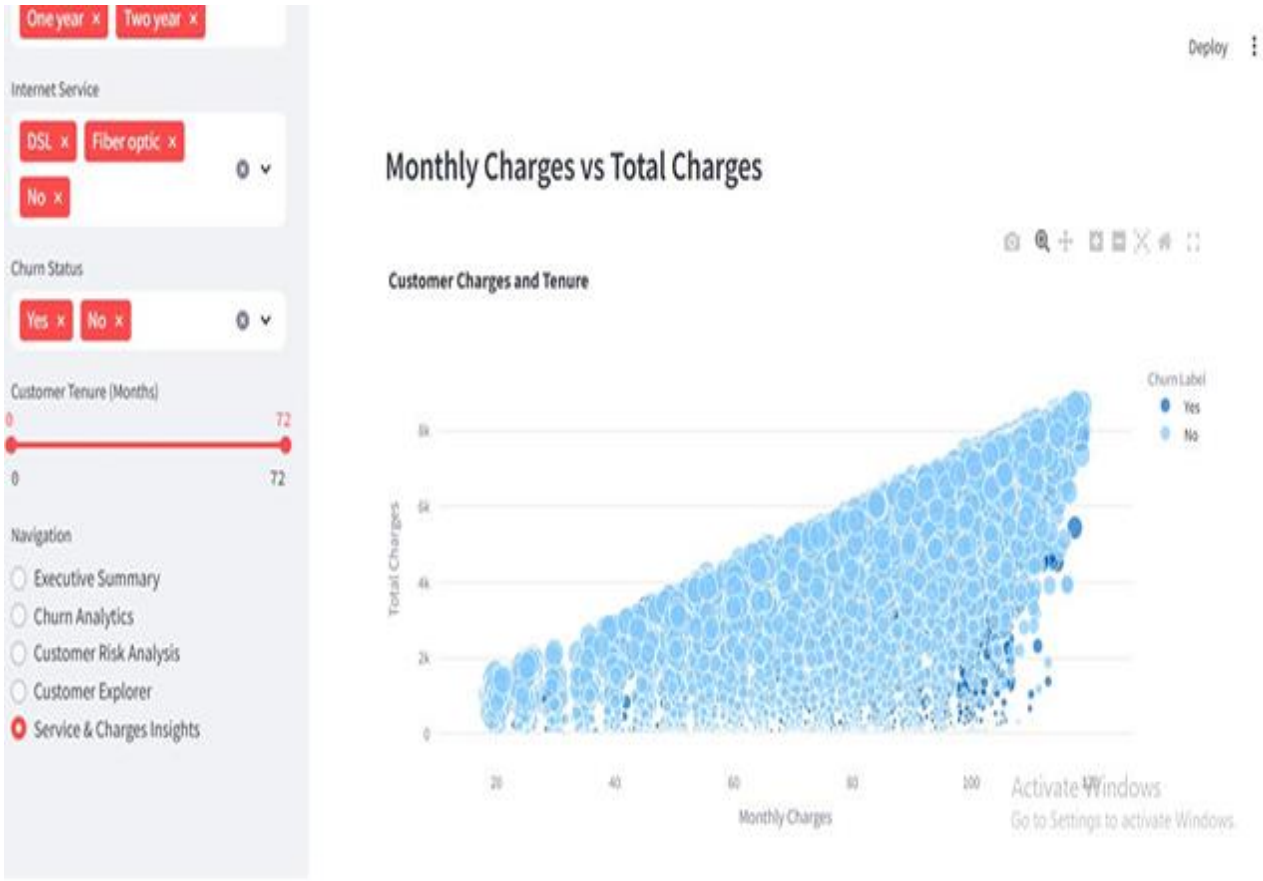




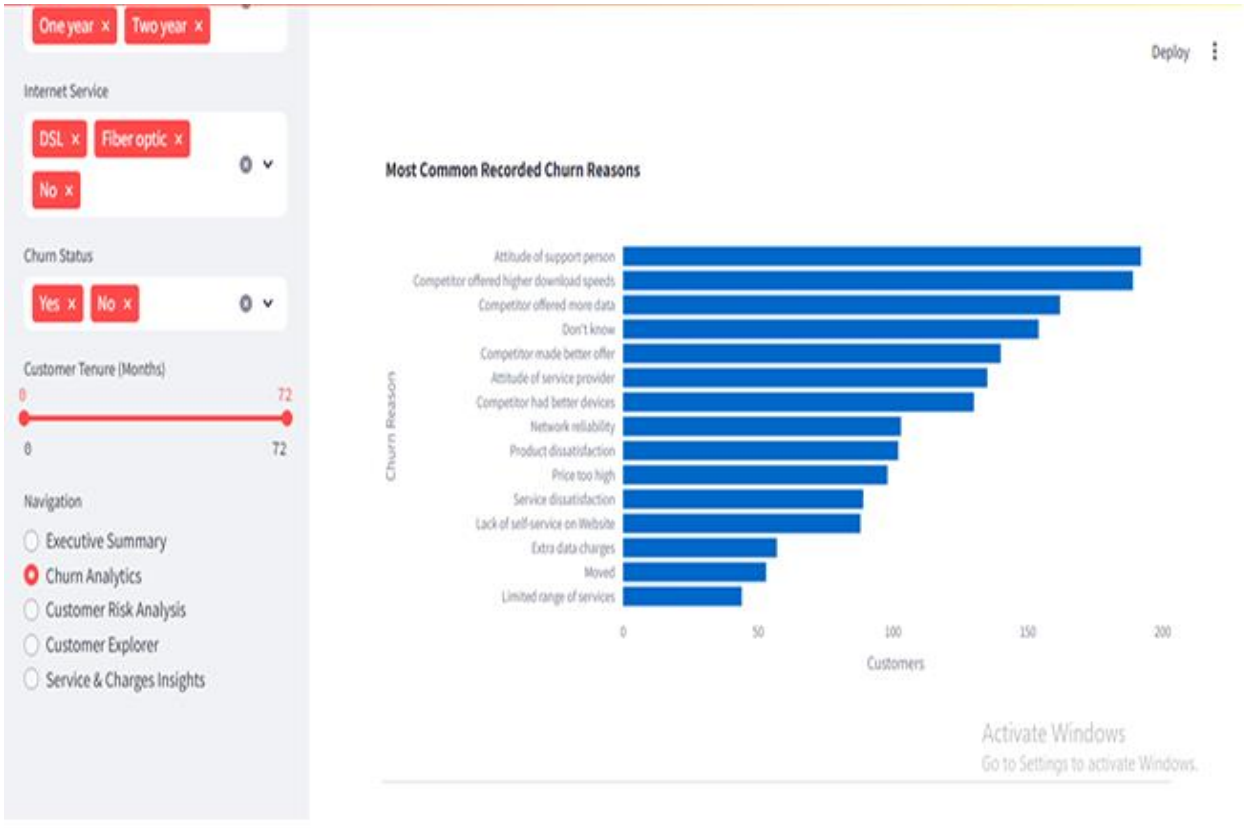

Deploy
Churn Intelligence
Customer Retention Analytics
Upload Telco Excel dataset (optional)
Drag and drop file here
Limit 200MB per file • XLSX
Browse files
Filters
Contract Type
Month-to-month
One year
Two year
Customer Churn Distribution
Churned vs Retained Customers
26.5%
73.5%
No
Yes
Churn Rate by Contract
Customer Churn by Contract Type
Churn Rate (%)
Month-to-month
One year
Two year
Contract
Activate Windows
Go to Settings to activate Windows.


Internet Service
DSL
Fiber optic
No
Churn Status
Yes
No
Customer Tenure (Months)
0
72
0
72
Navigation
Executive Summary
Churn Analytics
Customer Risk Analysis
Customer Explorer
Service & Charges Insights
Customer Churn Intelligence Platform
Telecommunications Customer Retention, Churn and Customer Lifetime Value Analytics
Showing 7,043 of 7,043 customers after applying sidebar filters.
Churn Analytics
Churn Rate by Internet Service
Churn Rate by Payment Method
Churn Rate (%)
Activate Windows
Go to Settings to activate Windows.

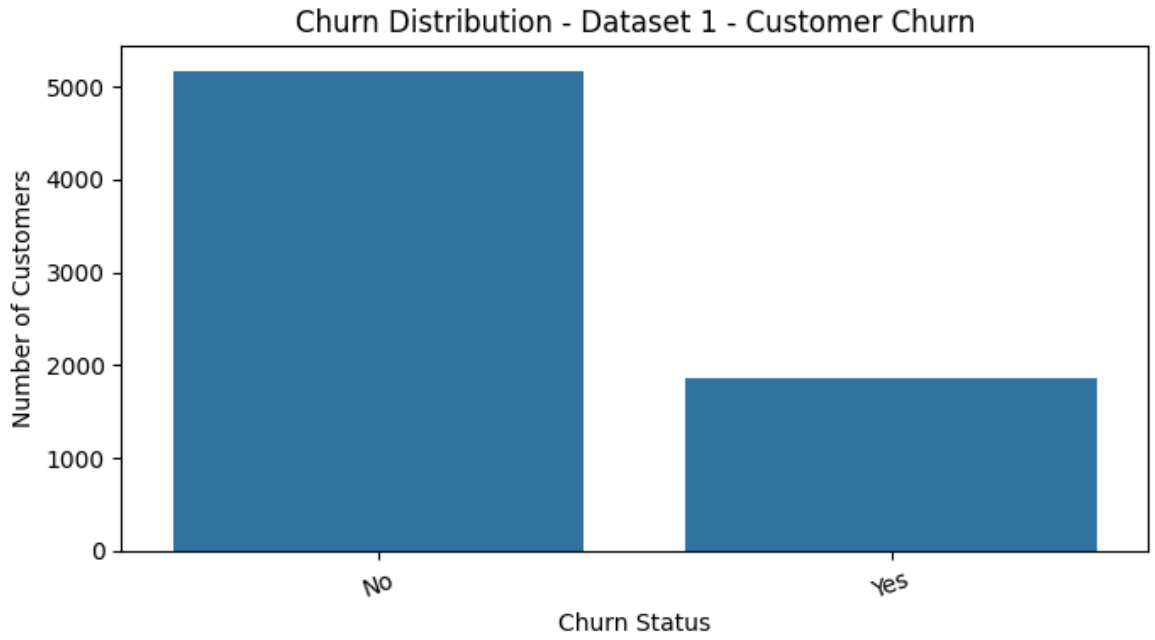

One year
Two year
Internet Service
DSL
Fiber optic
No
Churn Status
Yes
No
Customer Tenure (Months)
0
72
0
72
Navigation
Executive Summary
Churn Analytics
Customer Risk Analysis
Customer Explorer
Service & Charges Insights
Deploy
Monthly Charges by Churn Status
Monthly Charges
Churn Label
Yes
No
Tenure by Churn Status
Tenure Months
Churn Label
Yes
No
Activate Windows

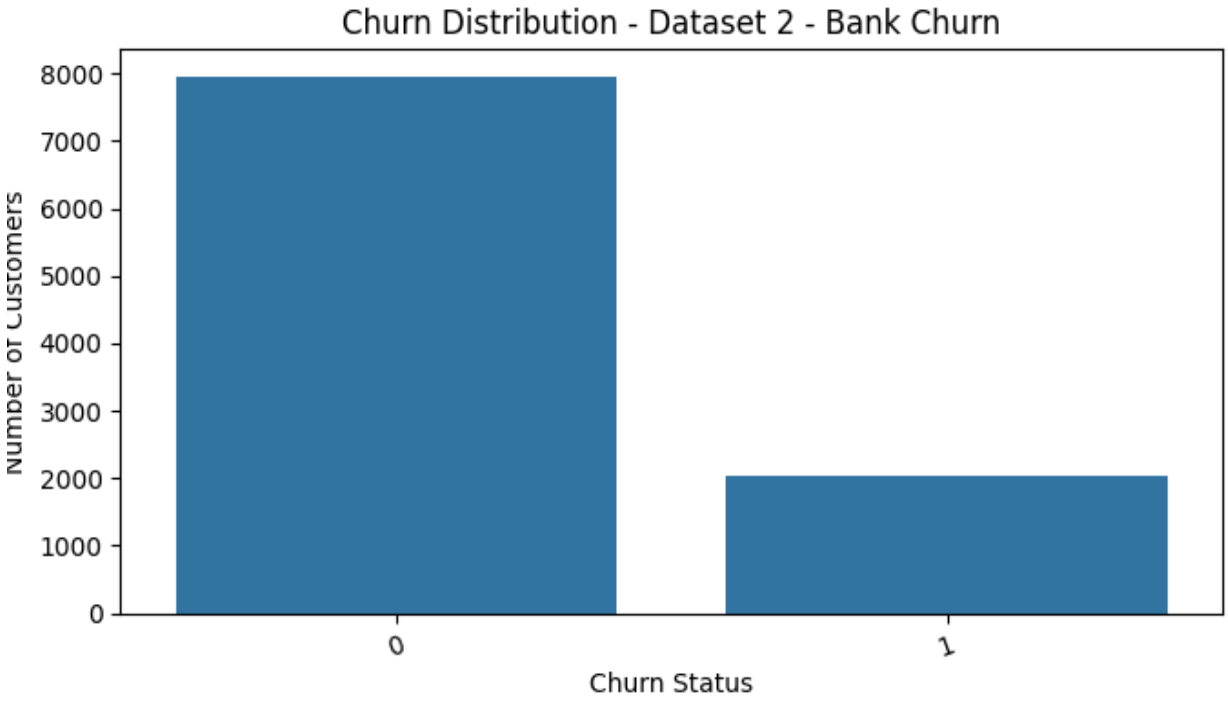

One year
Two year
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Customer Explorer
Service & Charges Insights
Deploy
Monthly Charges vs Total Charges
Customer Charges and Tenure
Churn Label
Yes
No
Total Charges
Monthly Charges
Activate Windows
Go to Settings to activate Windows

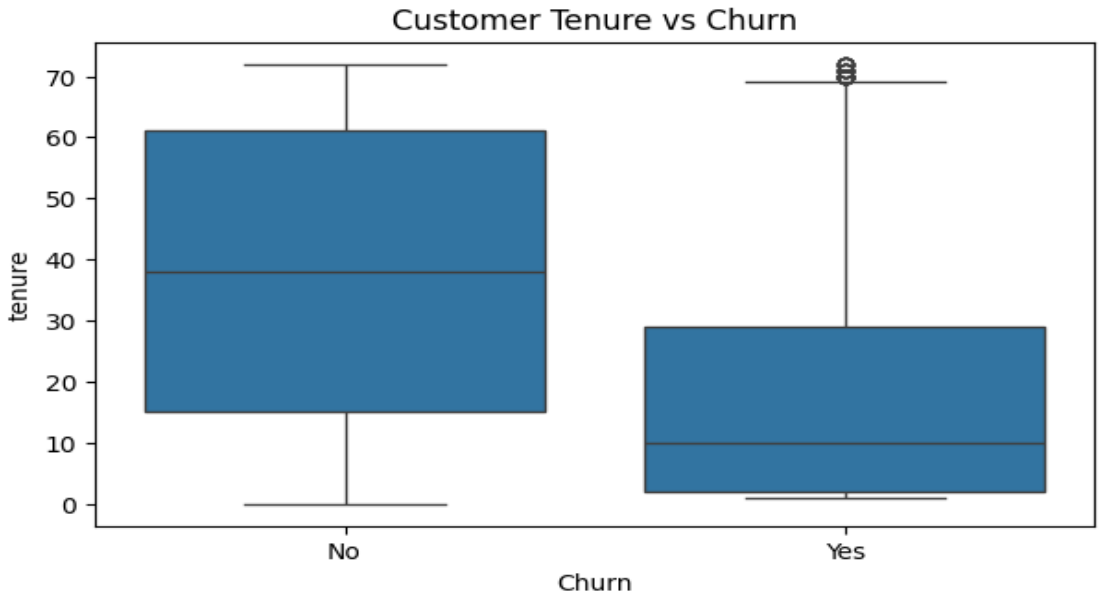

One year
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Churn Analytics
Customer Risk Analysis
Customer Explorer
Service & Charges Insights
Deploy
Customer Lifetime Value by Contract
Average CLTV by Contract Type
Average_CLTV
Contract
Month-to-month
One year
Two year
Activate Windows
Go to Settings to activate Windows.

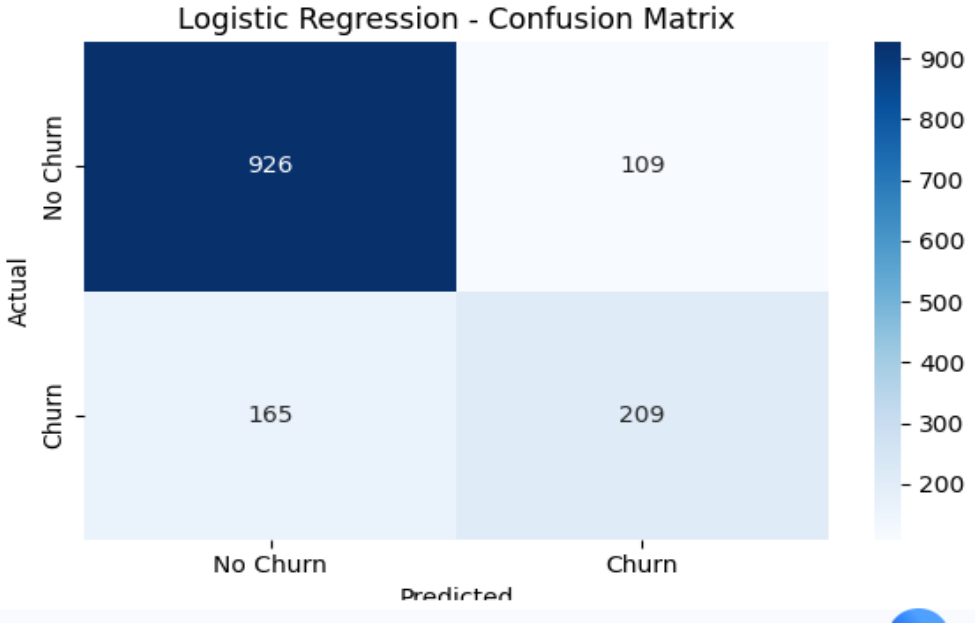

One year
Two year
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Churn Analytics
Customer Risk Analysis
Customer Explorer
Service & Charges Insights
Deploy
Most Common Recorded Churn Reasons
Churn Reason
Attitude of support person
Competitor offered higher download speeds
Competitor offered more data
Don't know
Competitor made better offer
Attitude of service provider
Competitor had better devices
Network reliability
Product dissatisfaction
Price too high
Service dissatisfaction
Lack of self-service on Website
Extra data charges
Moved
Limited range of services
Customers
Activate Windows
Go to Settings to activate Windows.

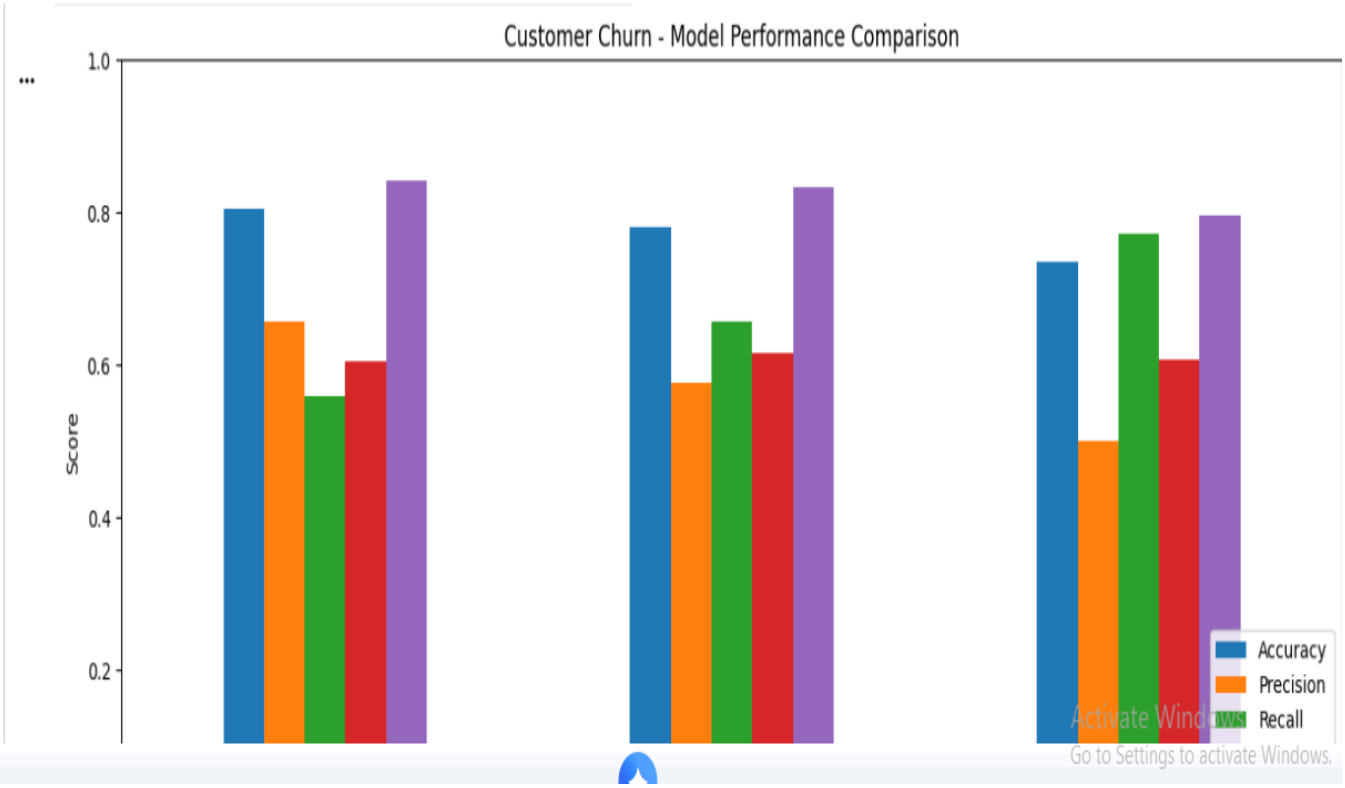
Machine Learning Models:






Customer Tenure vs Churn
tenure
70
60
50
40
30
20
10
0
No
Yes
Churn


Logistic Regression - Confusion Matrix
Actual
No Churn
Churn
926
109
165
209
No Churn
Churn
Predicted
900
800
700
600
500
400
300
200

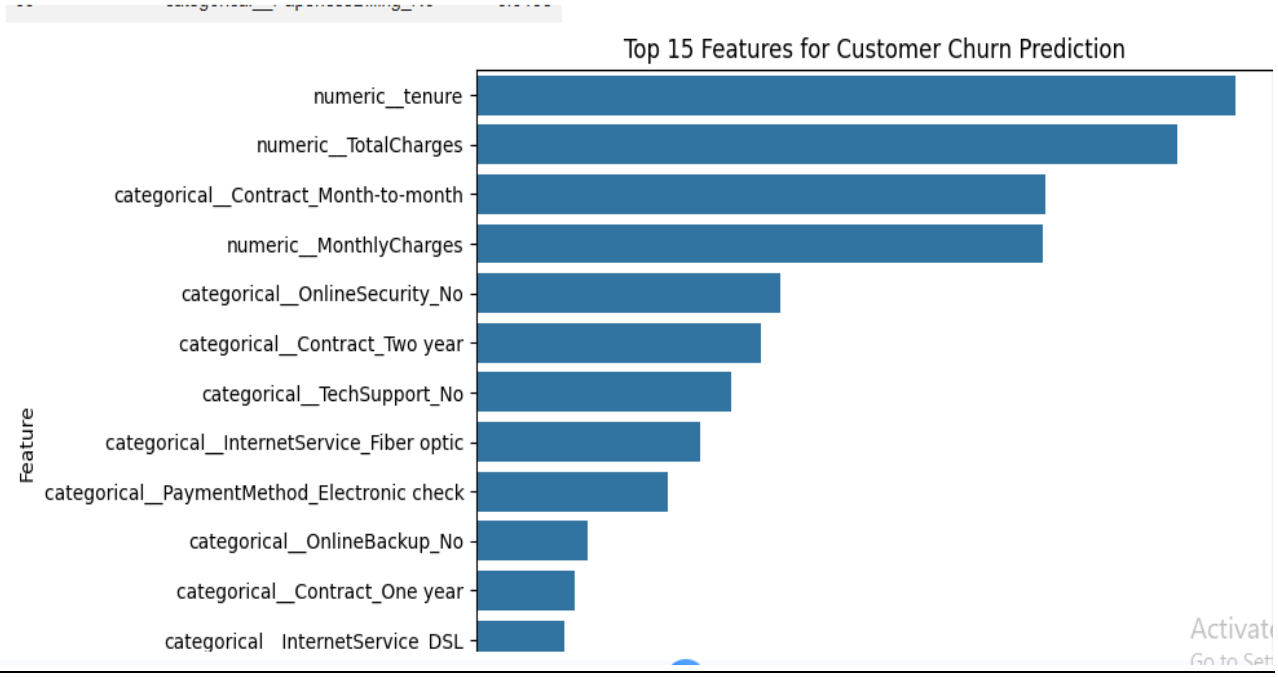


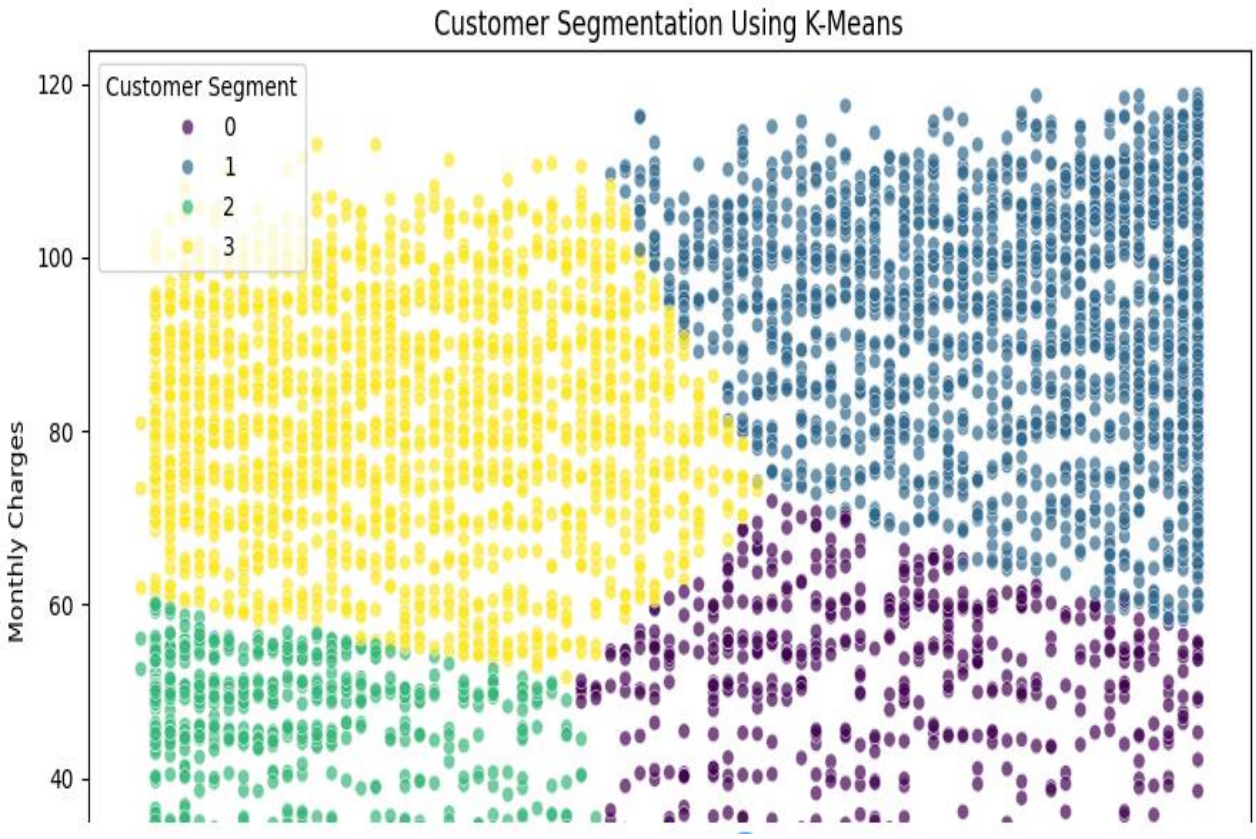
XGBoost Results

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	XGBoost	0.8055	0.6667	0.5348	0.5935	0.8445

Updated Model Comparison

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	XGBoost	0.8055	0.6667	0.5348	0.5935	0.8445
1	Logistic Regression	0.8055	0.6572	0.5588	0.6040	0.8419
2	Random Forest	0.7814	0.5775	0.6578	0.6150	0.8336
3	Decision Tree	0.7353	0.5009	0.7727	0.6078	0.7970


Top 15 Features for Customer Churn Prediction
numeric__tenure
numeric__TotalCharges
categorical__Contract_Month-to-month
numeric__MonthlyCharges
categorical__OnlineSecurity_No
categorical__Contract_Two year
categorical__TechSupport_No
categorical__InternetService_Fiber optic
categorical__PaymentMethod_Electronic check
categorical__OnlineBackup_No
categorical__Contract_One year
categorical InternetService DSL
Feature


Customer Segmentation Using K-Means
Customer Segment
0
1
2
3
120
100
80
60
40
Monthly Charges

Project Conclusion:

- ✓ Customer Churn Analysis – Successfully analyzed customer churn patterns, retention trends, and customer behavior using multiple datasets.
- ✓ Data Preprocessing – Performed data cleaning, transformation, and exploratory data analysis using Python, Pandas, and NumPy.
- ✓ Machine Learning Models – Developed and evaluated customer churn classification models to analyze customer attrition.
- ✓ Model Performance Evaluation – Analyzed machine learning performance using accuracy, precision, recall, and F1-score metrics.

Key Learnings – Customer Churn Analysis:

- ❖ Learned to identify customers who are likely to leave a service.
- ❖ Understood the major factors influencing customer churn.
- ❖ Learned to calculate churn rate and customer retention rate.
- ❖ Gained experience in analyzing churn data using Power BI dashboards.
- ❖ Understood how machine learning can help predict customer churn and improve customer retention.
- ❖ Learned to analyze customer behavior and usage patterns.
- ❖ Understood the relationship between monthly charges and customer churn.
- ❖ Learned to segment customers based on their characteristics.
- ❖ Gained knowledge of evaluating machine learning model performance.
- ❖ Learned to identify high-risk customer groups.
- ❖ Understood how data-driven insights can help businesses reduce customer churn.